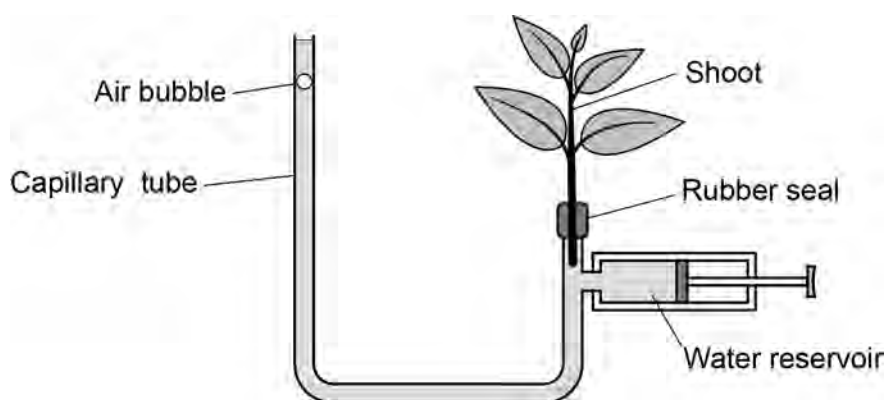


0 6

A student used a potometer to measure the movement of water through the shoot of a plant. The potometer is shown in **Figure 5**. As water is lost from the shoot, it is replaced by water from the capillary tube.

Figure 5



0 6 . 1

In one experiment, the air bubble moved 7.5 mm in 15 minutes. The diameter of the capillary tube was 1.0 mm.

Calculate the rate of water uptake by the shoot in this experiment.

Give your answer in mm^3 per hour. Show your working.
(The area of a circle is found using the formula, $\text{area} = \pi r^2$)

[2 marks]

_____ $\text{mm}^3 \text{ hour}^{-1}$

0 6 . 2

The student wanted to determine the rate of water loss per mm^2 of surface area of the leaves of the shoot in **Figure 5**.

Outline a method she could have used to find this rate. You should assume that all water loss from the shoot is from the leaves.

[3 marks]

0 6 . 3

The rate of water movement through a shoot in a potometer may not be the same as the rate of water movement through the shoot of a whole plant.

Suggest **one** reason why.

[1 mark]

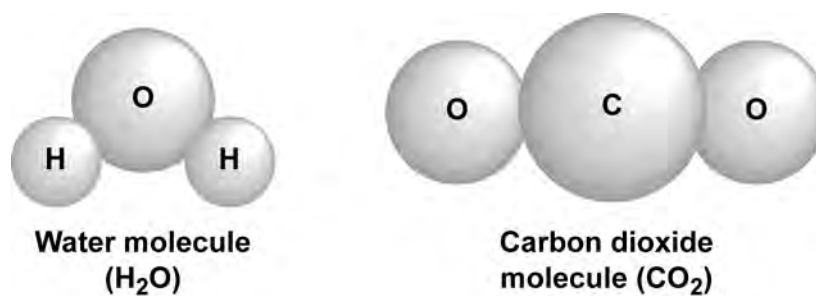
[Extra space]

0 6 . 4

Aquaporins are channel proteins that allow the diffusion of water across membranes. One type of aquaporin, called PIP1, can also transport carbon dioxide molecules across membranes.

Figure 6 shows the structure of a water molecule and of a carbon dioxide molecule. They are drawn to the same scale.

Figure 6



Suggest **two** reasons why water molecules **and** carbon dioxide molecules can both pass through PIP1.

[2 marks]

1 _____

[Extra space] _____

2 _____

[Extra space] _____

0 6 . 5

The scientists first produced transgenic poplar trees. These trees all had a length of foreign DNA inserted into them. This DNA led to the production of single-stranded RNA that specifically inhibited expression of the gene for PIP1.

The scientists then measured the difference in the amount of PIP1 in leaves of transgenic poplars and in leaves of wild type poplars without the foreign DNA. The amount of PIP1 in the transgenic poplars was approximately 15% of that in the wild type poplars.

Using this information, what can you conclude about the effect of the foreign DNA in the transgenic poplar trees?

[3 marks]

0 6 . 6

The transgenic poplars still produced some PIP1.
Suggest why.

[1 mark]

Question 6 continues on the next page

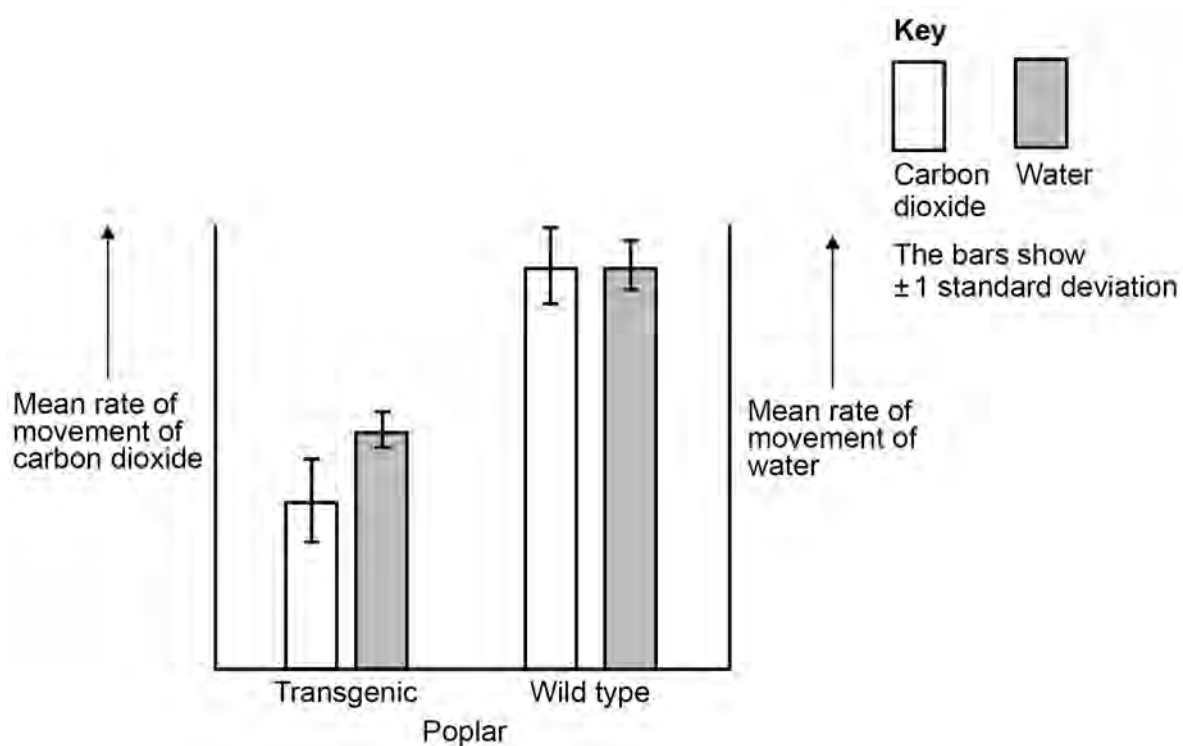
0 6 . 7

The scientists investigated the importance of PIP1 in the movement of water and carbon dioxide through the tissues of leaves of poplar trees.

They measured the mean rates of movement of carbon dioxide and water through the tissues of leaves of transgenic poplars and through the tissues of leaves of wild type poplars.

Their results are shown in **Figure 7**.

Figure 7



Using only **Figure 7**, evaluate the importance of PIP1 in the movement of carbon dioxide and water through leaves of poplar trees.

[3 marks]

1

Turn over for the next question

Turn over ►

Section B
Answer **one** question.